

FINAL YEAR PROJECT

Smart Solar Load Priority System

Keeping the important things on
when the sun goes down

[Your name] · Supervisor: [Supervisor's name] · [Department, university]

THE PROBLEM

When a solar battery runs flat, the whole house goes dark at once, security light included.

The battery charges by day and drains at night. If everything stays on, nothing is saved for the things that matter.

The idea: switch off the least important things first

18 hours

of security light and phone charging from the last stretch of charge, with this system

3.5 hours

from the same charge if every appliance is left on

Estimated from the load schedule for a 200 Ah, 48 V battery; not yet measured.

Every appliance goes in one of three groups

Critical

Would someone get hurt, or something spoil?

Security light, corridor light, phone and radio charging

Green light in the demo

Essential

Harder to live without, but nobody is harmed?

Ceiling fan, room lighting, refrigerator

Yellow light in the demo

Non-essential

Comfort, entertainment or decoration?

Decorative lights, TV, toilet light, extra bulbs in a lit room

Red light in the demo

Groups are decided by what happens if it goes off, not by how much power it uses

At home

Kitchen fridge = Essential

Food keeps for a few hours



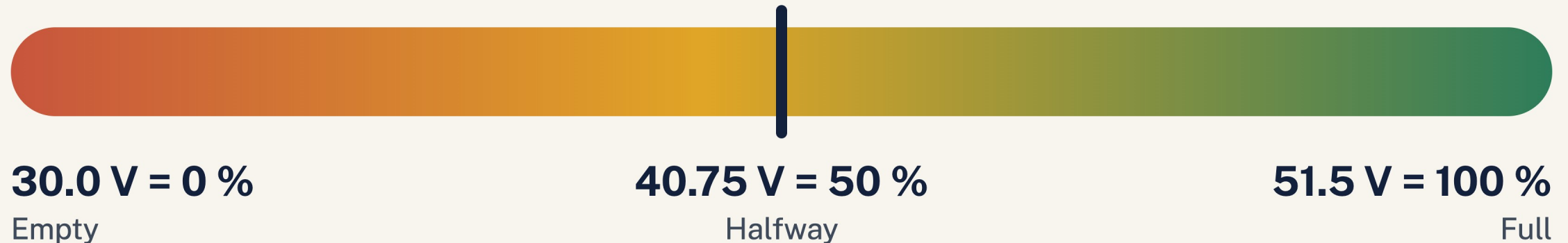
In a hospital

Vaccine fridge = Critical

Stock is lost within hours

It reads the battery like a fuel gauge

Only part of a battery's voltage range is safe to use. The controller turns that usable portion into a percentage.



It also **allows for voltage sag** when a heavy load starts, and **averages 50 readings a second**, so one noisy reading can't flip a switch.

As the battery drains, one group goes off at a time

100–66 %

Everything runs · 12.8 A

Critical · Essential · Non-ess.

66–33 %

Non-essential off · 8.3 A

Critical · Essential

33–10 %

Only Critical on · 2.5 A

Critical

Below 10 %

All off: battery protected · 0 A

Nothing: waiting for sun

When the sun returns, they come back in reverse order: most important first.

Like a thermostat, it doesn't flick on and off

The first version switched off and on at one voltage. The load pulled the voltage down, it switched off, the voltage recovered, it switched on again: thousands of times a second.

A gap between off and on

Off at 66 %, back on only at 73 %

Wait before acting

A brief dip is ignored; switching back waits longer

A rest period per switch

Each relay has a minimum time between changes

One switch at a time

Two groups never switch at the same instant

It also acts as a smart circuit breaker

When too much current is drawn

Above 20 A for a moment: Essential and Non-essential go off.
A motor's start-up surge doesn't set it off.

Above 26 A: off instantly.

Critical stays on. Too many appliances is not a flat battery.

Three trips in a row = a real fault. It stops retrying until reset.

What the buzzer means

Sound	Meaning
Fast beeping	Overload
Slow toll	Battery nearly empty
Chirp	Below 20 %
Silent	All normal

Live demo: draining the battery on command

Type	You will see	Because
soc 100	All three lights on, fan spinning	Full battery: everything runs
soc 60	Red light goes out	Below 66 %
soc 30	Yellow light out, fan stops	Below 33 %
soc 5	Green light out, buzzer tolls	Below 10 %: protect the battery
soc 100	Groups return one by one	Most important first

The screen shows volts, amps and battery %, e.g. **44.2V 6.4A 78%**, and which groups are on.

Built from cheap parts you can buy locally

~~₦~~24,000

for the control electronics

About ₦47,000 with a bench model to demonstrate on

Part	What it does
ESP32 board	The brain: reads and decides
16×2 LCD screen	Shows volts, amps, battery %
4-channel relay board	The switches for each group
ACS712 sensor	Measures current drawn
48 V to 5 V converter	Powers the electronics
Buzzer, LEDs, resistors	Alarm and indicator lights

Indicative prices from Alaba and Computer Village; confirm before quoting in the report.

Every change traces back to the supervisor's review

The supervisor asked for	Before	After
A real battery's usable portion	Fixed 11, 12, 13 V cut-offs	30.0–51.5 V range, decisions in %
Critical, essential, non-essential	High, Medium, Low, no appliances	Three groups with real appliances
See each load go off at its %	Screen showed raw volts	Screen shows %, every switch logged
Cheap, buyable parts	Never costed	About ₦24,000, all local
More than LEDs: add a motor	Three indicator lights	A fan on the Essential relay

Fixing these also caught the on-off flicker, and a wiring mistake that left the Critical light on permanently.

Honest limits, and what could come next

Count the charge

Voltage is a rough gauge. Counting current in and out is more accurate.

Measure the battery

Curves come from published figures; drain the real one and log it.

Watch the sun

40 % at 7 a.m. is not 40 % at 6 p.m.
Measure the panel too.

Time and motion

Toilet light on a motion sensor (about ₦1,500); security light at night only.

Phone dashboard

The board's Wi-Fi is unused; no extra parts needed.

Manual override

Let the user force a group on or off, with a timeout.

Thank you. Questions?

1. Three groups, decided by consequence
2. Switched off by battery %, least important first
3. About ₦24,000 of locally available parts